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Impact of burnout, job and life satisfaction, and life orientation on rationing of nursing care: a cross-sectional observational study

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Abstract

Background Rationing of nursing care leads to a decline in the quality of nursing care. With the current shortage of nursing staff being a major cause of care rationing, there is still a need for research to identify the factors influencing the level of care rationing. The aim was to evaluate the level of nursing care rationing among nursing staff, considering the type of rationed activities.

Methods The study included 795 nurses employed in 2018 at the University Clinical Hospital in Wrocław, who were directly responsible for patient care. Five standardized questionnaires were used to investigate the main variables – the Basel Extent of Rationing of Nursing Care (BERNCA-R), the Maslach Burnout Inventory (MBI), the Job Satisfaction Scale (JSS), the Satisfaction with Life Scale (SWLS), and the Life Orientation Test (LOT-R).

Results The average BERNCA-R score was 1.65 points (SD = 0.81), indicating a frequency of rationing between "never" and "rarely." The most rationed activity was the administration of prescribed medication or intravenous infusion at the appropriate time. Higher BERNCA-R scores (more frequent rationing) were statistically influenced by variables: lack of postgraduate education, working in closed healthcare, shift work, and a higher number of patients under care. The study showed an impact of life and job satisfaction, burnout on the BERNCA-R score. Independent predictors of a higher BERNCA-R score (more frequent rationing) were specialization, having 16–25, 26–35, or > 35 patients under care, a less pessimistic life orientation, higher emotional exhaustion, and greater depersonalization. Independent predictors of a lower BERNCA-R score (less frequent rationing) were professional experience of more than 20 years, open healthcare as the primary workplace, and "other" as the main workplace.

Conclusions Higher life and job satisfaction reduce the level of nursing care rationing. Conversely, rationing is more frequent with higher burnout levels, among those working in closed healthcare and shift work. An increase in the number of patients under care during shifts leads to increased burnout and more frequent rationing of nursing care.

Clinical trial number Not applicable.

Keywords Nursing staff, Nursing care rationing, Professional burnout, Life satisfaction, Job satisfaction, Life orientation

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Background

The concept of rationing nursing care and its synonyms (loss, inadequacy, delay, omission, circumvention, neglect, missed care, unsuccessful care, hidden rationed care) has a history spanning several years. Rationing nursing care is defined as an omission, delay, or unfinished care, and it is also referred to as an error of omission. This concept's expansion can be found in the article by Kalisch et al. [1, 2]. Through the utilization of MISSCARE, these authors demonstrated that the lack of care was due to work resources (85%), material resources (56%), and communication (38%). The activities neglected were as follows: assessment was missed by 44% of respondents, interventions and basic health care were overlooked by 73% of nurses, and planning by 71% of study participants. The results of this study conclude that a large percentage of patients were at risk due to the lack of care or omission error. The most commonly cited reasons for care omission among work resources were: unexpected deterioration in the patient's health, acute cases, low staffing levels, and an insufficient number of support staff. These reasons were reported by 80% of the surveyed nursing staff.

The nursing environment in Poland has long been striving to establish mechanisms to ensure appropriate nursing staffing, which is the starting point for providing nursing care according to patient demand. The consequences of nursing shortages have affected the nursing professional group itself as the work has become increasingly challenging. This situation has also affected society at large due to the growing need for nursing care [3]. Limited interest in the profession among younger generations, nurses' emigration, especially those with higher education and foreign language skills, pose challenges for the nursing profession and the Polish state [4]. According to the Supreme Chamber of Nurses and Midwives, the number of nurses and midwives to whom certificates were issued for professional qualifications recognition in EU countries from 2004 to 2016 was 19,953 [5], whereas the European Commission revealed that Polish nurses ranked 5th in the number of emigrants [6].

Among the reasons for rationing nursing care, B.J. Kalisch [7] lists 5 areas that affect each other, creating a causal chain leading to care delays/omissions: factors related to the nurse's work environment (staffing, material resources, relationships, teamwork, communication within the team), factors related to the care process, factors related to management and decision-making processes, nursing care - provided according to the plan, delayed or omitted, and patient outcomes. The most frequently cited reason in the literature for rationing nursing care is staffing shortages. In a cross-sectional study involving nearly 2400 nurses in Australia, Hegney et al. [8] showed that 20 to 40% of surveyed nurses reported not being able to provide the required care due to insufficient staff and inadequate skills, resulting in excessive workloads and rationing of nursing care. Blackam et al. [9] identified 8 causes of rationing nursing care: staff availability, shift work, nurse staffing allocation – resource allocation, team communication, workload, predictability of workload, nurses' satisfaction with their current work, and intentions to change jobs. It should be emphasized that the greater the deficiencies in human and material resources, job dissatisfaction, and intent to leave the job, the more frequent the care rationing.

In the context of rationing of nursing care, nurse burnout emerges as a significant occupational hazard affecting not just nurses but also patients, healthcare organizations, and society at large. Nurse burnout is associated with decreased safety and quality of care, reduced patient satisfaction, and lowered organizational commitment and productivity among nurses. Traditionally considered an individual issue, burnout should be reframed as an organizational and collective phenomenon to effectively address its impact on the rationing of nursing care [10]. Moreover, recent studies conformed a strong correlation between occupational burnout and depression among nurses, which may contribute to the increased rationing of care due to diminished nurse well-being and capacity [11].

Considering the insufficient number of studies concerning the influence of factors related to nurses on rationing nursing care, the aim of this study was to evaluate the level and types of nursing care rationing among nursing staff and to explore associations between care rationing and selected variables, including sociodemographic characteristics, work-related factors, professional burnout, job satisfaction, life satisfaction, and life orientation among nurses.

Methods

Study desing and settings

This study utilized a cross-sectional, observational design to investigate factors related to nursing care rationing, burnout, job satisfaction, life satisfaction, and life orientation among nurses. The study was conducted at the University Clinical Hospital in Wrocław, Poland, one of the largest healthcare institutions in the region. Data were collected in 2018 from a group of 832 registered nurses who were actively engaged in providing direct patient care. The participants were selected using a purposive sampling method, targeting nurses who met the eligibility criteria for the study. These criteria included current employment at the hospital and direct involvement in patient care. Nurses working in managerial or supervisory positions, such as head nurses and coordinators, were excluded from participation to focus on frontline nursing staff who are more likely to experience the

direct effects of care rationing and burnout. This exclusion criterion was applied to ensure the study reflected the experiences of staff nurses rather than those with administrative responsibilities.

Ethical approval

The research project was approved by the Bioethics Committee of Wroclaw Medical University (approval no. KB-691/2017). The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines were followed. Clinical trial number is not applicable for this study due its character of original research instead clinical trial. The study was designed and conducted in accordance with the Declaration of Helsinki (1964) and Good Clinical Practice (GCP) guidelines. The researchers took precautions to ensure that participation in the study did not interfere with the nurses' routine duties or workload, thereby minimizing any additional burden on the participants. The voluntary nature of the study was emphasized, and participants were reassured that their responses would remain confidential and anonymous. To further uphold the ethical standards, no financial or other incentives were offered for participation, thus reducing any undue influence on the responses provided.

Data collection

The data collection process involved distributing self-administered paper questionnaires to nurses during their work shifts. The nursing staff were informed about the purpose and significance of the study, and the importance of their participation in understanding and improving nursing care practices. To avoid any disruption to patient care duties, participants were allowed to complete the surveys at their convenience, either during their breaks or at the end of their shifts. Participants were assured of the voluntary nature of their participation and were provided with clear instructions for completing the questionnaire. Sufficient time was allotted to each nurse to ensure that they could thoughtfully and accurately respond to each item. The estimated time required to complete the entire survey was approximately 20–30 min, depending on the individual participant's pace. To preserve anonymity and ensure that participants felt comfortable providing honest and accurate responses, no personal identifiers were included in the questionnaires. After completing the survey, participants placed their questionnaires in sealed envelopes and submitted them anonymously in designated collection boxes located in convenient areas within the hospital, such as break rooms or nursing stations. The research team retrieved the envelopes daily to ensure the security and confidentiality of the data collected. The collection process was designed to maximize response rates while minimizing any burden on the nursing staff. Reminders were given periodically to encourage participation, and nurses were provided with multiple opportunities to complete and return the questionnaire during the data collection period. This approach resulted in a high response rate, with 795 out of 832 eligible nurses completing the survey, yielding a completion rate of 95.55%.

Outcome measures

A custom questionnaire was used to collect sociodemographic data and work-related variables. Standard research tools were employed to evaluate various study outcomes: the Basel Extent of Rationing of Nursing Care (BERNCA-R), the Maslach Burnout Inventory (MBI), the Job Satisfaction Scale (JSS), the Satisfaction with Life Scale (SWLS), and the Life Orientation Test (LOT-R).

The BERNCA-R questionnaire was utilized to measure the extent to which nursing care was rationed over the preceding seven days. This tool consists of 32 questions that assess the frequency with which nurses were unable to perform specific patient care tasks. Respondents rated each item on a 5-point Likert scale ranging from 0 (never) to 4 (often). A higher score on the BERNCA-R scale reflects a greater frequency of rationing nursing care tasks. This instrument has been validated and widely used in nursing research to evaluate implicit care rationing, providing reliable insights into which aspects of patient care are most frequently omitted due to constraints such as time or resource limitations [12, 13].

The MBI was employed to evaluate burnout among nurses, a phenomenon characterized by emotional exhaustion, depersonalization, and a diminished sense of personal accomplishment. The tool comprises three subscales: Emotional exhaustion (EE): Assesses feelings of being emotionally overextended and exhausted by one's work. Depersonalization (DEP): Measures an unfeeling and impersonal response towards recipients of one's care or service. Personal accomplishment (PA): Reflects feelings of competence and successful achievement in one's work with people. Each subscale generates a score ranging from 0 to 100 points, with higher scores in emotional exhaustion and depersonalization indicating higher levels of burnout, while lower scores in personal accomplishment indicate a reduced sense of professional achievement [14–16].

The JSS is a commonly used instrument for assessing overall job satisfaction among professionals. It consists of 7 items, each rated on a 5-point scale, with the total score ranging from 5 to 35 points. A higher score on the JSS signifies greater job satisfaction, whereas lower scores indicate dissatisfaction with the work environment or conditions. The tool is designed to capture both intrinsic (personal fulfillment from the work itself) and extrinsic

(satisfaction with work conditions) aspects of job satisfaction [17].

The SWLS was employed to assess the respondents' overall life satisfaction. This scale consists of 5 items, with responses measured on a 7-point scale ranging from 1 (strongly disagree) to 7 (strongly agree). The total score, therefore, ranges from 5 to 35 points, where higher scores indicate a greater level of satisfaction with life. This tool evaluates the cognitive component of subjective well-being and is widely used in research to gauge life satisfaction independently of other emotional states [18, 19].

The LOT-R is a 10-item questionnaire designed to assess dispositional optimism, which is the general expectation that good things will happen in the future. Of the 10 items, 6 contribute directly to the optimism score, while the remaining 4 are filler items. Responses are measured on a 5-point Likert scale ranging from 0 (strongly disagree) to 4 (strongly agree), yielding a total score between 0 and 24 points. A higher score reflects a more optimistic outlook on life, while lower scores suggest a more pessimistic orientation [20, 21].

Sample size

The participants were selected using a purposive sampling method, targeting nurses who met the eligibility criteria for the study. A retrospective power analysis was conducted using G*Power software (version 3.1.9.7) for linear multiple regression analysis with a small to moderate effect size ($f^2 = 0.05$), alpha of 0.05, power of 0.80, and up to 15 predictor variables. The required total sample size was 636. Considering an anticipated dropout or incomplete data rate of approximately 20%, the adjusted sample size required was 795 nurses. Thus, the final sample size of 795 participants ensured adequate statistical power and compensated for potential data loss.

Statistical analysis

The research data analysis was carried out using R software, version 4.0.5 (R Project for Statistical Computing, Vienna, Austria) [22]. Descriptive statistics were first calculated for all demographic and study variables, providing an overview of the sample characteristics (e.g., age, gender, professional experience, etc.). Continuous variables were reported as means with standard deviations (SD), and categorical variables as frequencies with percentages. To compare the quantitative results between two groups, non-parametric tests were employed due to the non-normal distribution of the data, as assessed by the Shapiro-Wilk test. The Mann-Whitney U test was applied for two-group comparisons, and the Kruskal-Wallis test was used for comparisons involving more than two groups. For correlation analysis between continuous or ordinal variables, Spearman's rank correlation coefficient was calculated. This non-parametric measure was chosen to assess the strength and direction of the relationship between variables such as burnout, life satisfaction, job satisfaction, and the rationing of nursing care (BERNCA-R score). Additionally, a multivariate regression analysis was performed to identify independent predictors of nursing care rationing. Multiple linear regression was used to model the relationship between the dependent variable (nursing care rationing as measured by the BERNCA-R score) and several independent variables, including professional experience, number of patients under care, work setting, and levels of burnout. The variables included in the regression model were selected based on their statistical significance in the univariate analyses ($p < 0.05$) and their theoretical relevance to the study's objectives. The regression coefficients (β) were calculated, providing estimates of the expected change in the BERNCA-R score per unit increase in each predictor variable. Confidence intervals (95% CI) and p-values were also reported to assess the precision and statistical significance of the estimates. The level of statistical significance for all tests was set at $p < 0.05$. The goodness-of-fit of the regression model was evaluated using the R-squared (R^2) statistic, which indicated the proportion of variance in the dependent variable explained by the independent variables. All analyses were two-tailed and interpreted with caution, considering potential confounding factors that may not have been fully accounted for in the model.

Results

Participant demographics

The study group consisted of 795 nurses. Among the participants, the largest segment was female, accounting for 763 (95.97%), while only 16 (2.01%) were male. The largest group within the study were individuals aged 41–50 years – 331 individuals (41.64%), followed by the age group of 51–60 years – 223 individuals (28.05%). Those aged 23–40 years collectively represented 27.68% of the total respondents (220 individuals). Regarding educational background, the highest number of respondents, 328 individuals (41.26%), indicated having completed Medical High School, constituting the largest group among the participants. The next significant group was those with a bachelor's degree in nursing – 226 individuals (28.43%). Those with a master's degree in nursing were 142 individuals (17.86%), and 89 individuals (11.19%) had completed medical studies. Regarding postgraduate education, participants mostly engaged in specialized courses, as indicated by 429 individuals (53.96%). This was followed by qualification courses – 339 individuals (42.64%) and nursing specialization courses – 228 individuals (28.68%).

A significant portion, comprising over half of the surveyed individuals, possessed work experience exceeding

Table 1 Sociodemographic and professional analysis of the studied group ($n = 795$)

Parameter		*N* (%)
Age	23–30 years	134 (16.86%)
	31–40 years	86 (10.82%)
	41–50 years	331 (41.64%)
	51–60 years	223 (28.05%)
	> 60 years	19 (2.39%)
	Missing data	2 (0.25%)
Sex	Female	763 (95.97%)
	Male	16 (2.01%)
	Missing data	16 (2.01%)
Education	Medical secondary school	328 (41.26%)
	Medical study	89 (11.19%)
	Bachelor of Science in Nursing	226 (28.43%)
	Master of Science in Nur	142 (17.86%)
	Other	3 (0.38%)
	Missing data	7 (0.88%)
Professional seniority	< 5 years	143 (17.99%)
	6–10 years	61 (7.67%)
	11–15 years	40 (5.03%)
	16–20 years	82 (10.31%)
	> 20 years	457 (57.48%)
	Missing data	12 (1.51%)
Postgraduate education	Specialization	228 (28.68%)
	Qualification courses	339 (42.64%)
	Specialized courses	429 (53.96%)
	Other forms	84 (10.57%)
Number of jobs	1	453 (56.98%)
	2	234 (29.43%)
	3	46 (5.79%)
	4	17 (2.14%)
	> 5	16 (2.01%)
	Missing data	29 (3.65%)
Main place of work	Inpatient treatment	740 (93.08%)
	Outpatient treatment	38 (4.78%)
	School, orphanage, emergency care	9 (1.13%)
	Long-term care	4 (0.5%)
	Private care	1 (0.13%)
	Other	1 (0.13%)
	Missing data	2 (0.25%)
System of work	Single shift	126 (15.85%)
	Double shift	650 (81.76%)
	Other	14 (1.76%)
	Missing data	5 (0.63%)
Average number of patients under care	1–5 patients	111 (13.96%)
	6–15 patients	239 (30.06%)
	16–25 patients	210 (26.42%)
	26–35 patients	74 (9,31%)
	> 35 patients	159 (20.0%)
	Missing data	2 (0.25%)

20 years – 457 individuals (57.48%). The next segment comprised individuals with work experience up to 5 years – 143 individuals (17.99%), followed by those with 16–20 years of experience – 82 individuals (10.31%) and 6–10 years of experience – 61 individuals (7.67%). The fewest individuals had work experience ranging from 11 to 15 years – 40 individuals (5.03%). More than half of the respondents declared working in a single workplace – 453 individuals (56.98%). The remaining group of respondents (excluding those who did not provide responses – 29; 3.65%) included those with two or more workplaces – 313 individuals (39.37%). Specifically, 234 individuals (29.43%) mentioned having two workplaces, 46 individuals (5.79%) three workplaces, 17 individuals (2.14%) four workplaces, and 16 individuals (2.01%) mentioned having five or more workplaces. Closed healthcare settings were indicated as the main workplace by 740 individuals (93.08%), while 38 individuals (4.78%) mentioned other closed healthcare settings. The least number of respondents, totaling 15 individuals (1.89%), mentioned other primary workplaces, such as schools, children's homes, emergency care, long-term care, private corporate care, and other workplaces apart from those mentioned. Shift work was the most commonly selected work system, accounting for 81.76% of respondents (650 individuals). A single-shift system was reported by 126 individuals (15.85%). The average number of patients under care during a shift, according to the most commonly selected responses, were: 6–15 patients – 239 individuals (30.06%), 16–25 patients – 210 individuals (26.42%), over 35 patients – 159 individuals (20%), 1–5 patients – 111 individuals (13.96%), and 26–35 patients – 74 individuals (9.31%). These findings are summarized in Table 1.

Measured outcomes

It was shown that 300 out of 795 survey participants (37.74%) had a high sense of life satisfaction, 281 respondents (35.35%) had moderate life satisfaction, 197 respondents (24.78%) had low life satisfaction, and 17 respondents (2.14%) left the questionnaire unanswered. The average score for the JSS was 19.09 points, which equates to 3.82 points per question. Hence, the respondents were neither satisfied nor dissatisfied with their work. 282 out of 795 survey participants (35.47%) had a neutral orientation, 256 respondents (32.20%) were inclined toward pessimism, 232 respondents (29.18%) were inclined toward optimism, and 25 respondents (3.14%) left the questionnaire unanswered. The overall burnout index was an average of 36.98 points out of 100 possible (SD = 21.94), ranging from 0 to 95.83 points. Emotional exhaustion accounted for the highest degree of burnout among the respondents (average 43.79 points), followed by a lower degree of job dissatisfaction (average 37.75 points), and the least was depersonalization

(average 29.39 points). The above data is presented in Table 2.

BERNCA-R responses

The average total score of BERNCA-R was 1.65 points (SD = 0.81). Therefore, it can be concluded that the frequency of care rationing by the respondents lies between "never" and "rarely". The most frequently rationed care aspects (highest averages) were: administering prescribed medication or infusion at the right time (question 23), familiarizing oneself with the condition of individual patients and care plans at the beginning of the shift (question 29), providing emotional or psychosocial support to the patient (question 10), conversation with the patient or family (question 11), and monitoring the patient's condition as thoroughly as needed (question 19). The specific data is presented in Table 3.

Effect of demographics on rationing of nursing care

The results of the comparative analysis of the influence of specific variables on the level of nursing care rationing are presented in Table 4. The frequency of care rationing was significantly higher in the group without qualifying courses (1.71 ± 0.79 vs. 1.56 ± 0.83), without specialized courses (1.72 ± 0.79 vs. 1.59 ± 0.82), and in closed healthcare settings compared to open ones (1.69 ± 0.79 vs. 1.15 ± 0.76), where it was significantly higher than in "other" settings. The frequency of care rationing was significantly higher in shift work compared to other groups. Furthermore, the frequency of care rationing was significantly higher with 26–35 patients than with 1–5, 6–15, and 16–25 patients, and also higher with over 35 patients than with 1–5 and 6–15 patients. In addition, it was significantly higher with 6–15 and 16–25 patients than with 1–5 patients.

Relationship between life and jobsatisfaftion, life orientation and burnout and rationing of nursing care

The results of the correlation analysis showing the impact of specific variables on the level of nursing care rationing are presented in Table 5. A lower frequency of care rationing was significantly and negatively correlated with life satisfaction (r = -0.261). This indicates that higher job satisfaction is associated with lower care rationing frequency. Conversely, a higher frequency of care rationing was significantly and positively correlated with the overall score of the MBI questionnaire and its domains. This implies that a higher overall MBI score (indicating higher burnout), increased emotional exhaustion, greater depersonalization, and a stronger sense of job dissatisfaction are associated with higher care rationing frequency.

Independent predictors of rationing of nursing care

The multi-factor linear regression model revealed that the significant ($p < 0.05$) independent predictors of the BERNCA-R score are: Work experience in the profession > 20 years: The regression parameter is -0.485, indicating that it reduces the BERNCA-R score by an average of 0.485 points compared to work experience of up to 5 years. Specialization: The regression parameter is 0.144, implying that it raises the BERNCA-R score by an average of 0.144 points. Open healthcare as the main workplace: The regression parameter is -0.441, leading to an average reduction of 0.441 points in the BERNCA-R score compared to closed healthcare. "Other" main workplace: The regression parameter is -0.763, causing an average reduction of 0.763 points in the BERNCA-R score compared to closed healthcare. 16–25 patients under care: The regression parameter is 0.191, increasing the BERNCA-R score by an average of 0.191 points compared to 1–5 patients. 26–35 patients under care: The regression parameter is 0.528, increasing the BERNCA-R score by an average of 0.528 points compared to 1–5 patients. 35 patients under care: The regression parameter is 0.354, raising the BERNCA-R score by an average of 0.354 points compared to 1–5 patients. LOT-R: The regression parameter is 0.021, indicating that every "point of optimism" increases the BERNCA-R score by an average of 0.021 points. MBI: Emotional exhaustion: The regression parameter is 0.006, suggesting that every "point of emotional exhaustion" increases the BERNCA-R score by an average of 0.006 points. MBI: Depersonalization: The

Table 2 Results of the BERNCA-R, SWLS, JSS, LOT-R and MBI questionnaires

Questionnaires	*N*	Mean	SD	Median	Min	Max	Q1	Q3
BERNCA-R	795	1.65	0.81	1.56	0	4	1.09	2.12
SWLS	778	21.65	5.81	22	5	35	17	26
JSS	779	19.09	6.12	3.82	19	5	35	14
LOT-R	770	14.43	3.81	14	0	24	12	17
MBI	783	36.98	21.94	35.83	0	95.83	18.98	53.43
MBI: EE	785	43.79	30.07	44.44	0	100	22.22	66.67
MBI: DEP	783	29.39	27.77	20	0	100	0	40
MBI: PA	792	37.75	28.15	37.5	0	100	12.5	62.5

Abbreviations: MBI – Maslach Burnout Inventory General Survey; EE – emotional exhaustion; DEP – depersonalization; PA – lack of personal accomplishments; SWLS – Satisfaction with Life Scale; LOT-R – Life Orientation Test-Revised; JSS – Job Satisfaction Scale; BERNCA-R – Basel Extent of Rationing of Nursing Care revised

Table 3 Distribution of responses to individual questions of the BERNCA-R questionnaire

Question	There was no need (0)	Never (1)	Rarely (2)	Sometimes (3)	Often (4)	No answer	Average score
1. Sponge bath	26.67%	31.45%	19.87%	13.33%	7.04%	1.64%	1.42
2. Partial sponge bath	25.41%	34.84%	18.87%	15.09%	3.77%	2.01%	1.36
3. Skin care	17.74%	33.71%	22.64%	18.87%	5.16%	1.89%	1.59
4. Oral hygiene	19.37%	28.93%	23.27%	17.61%	8.43%	2.39%	1.66
5. Dental hygiene	21.51%	24.91%	22.64%	18.49%	9.94%	2.52%	1.7
6.Assist food intake	19.25%	34.34%	25.66%	12.70%	5.41%	2.64%	1.49
7. Mobilization	18.62%	19.37%	28.68%	21.01%	9.43%	2.89%	1.83
8. Change of the position	20.25%	28.30%	23.52%	17.86%	7.30%	2.77%	1.63
9. Change of the bed linen	15.47%	26.54%	29.81%	16.48%	9.69%	2.01%	1.78
10. Emotional and psychological support	12.83%	22.14%	28.55%	21.13%	12.96%	2.39%	1.99
11. Necessary conversation	12.58%	19.87%	31.82%	23.65%	9.81%	2.26%	1.98
12. Information about therapies	12.70%	30.82%	28.05%	19.12%	6.79%	2.52%	1.76
13. Continence training (diapers)	29,31%	22,14%	21,64%	16,60%	7,67%	2,64%	1,5
14. Continence training (insert catheter)	35.47%	28.93%	15.35%	12.96%	4.78%	2.52%	1.21
15. Activating or rehabilitating care	30.94%	13.46%	15.85%	17.74%	19.37%	2.64%	1.81
16. Education and training	30.31%	19.75%	21.51%	18.24%	7.67%	2.52%	1.52
17. Preparation for discharge	23.27%	25.28%	24.65%	18.74%	6.67%	1.38%	1.6
18. Monitoring patients as described by physician	11.95%	30.06%	27.42%	19.50%	9.94%	1.13%	1.85
19. Monitoring patients as the nurse felt necessary	9.94%	27.80%	28.68%	20.63%	11.32%	1.64%	1.96
20. Monitoring of confused patients and use of restrains	34.97%	23.02%	16.60%	16.48%	7.30%	1.64%	1.37
21. Monitoring of confused patients and use of sedatives	32.96%	28.93%	15.47%	15.60%	5.53%	1.51%	1.31
22. Delay in measure because of a physician delay	20.00%	32.20%	24.28%	16.60%	5.28%	1.64%	1.54
23. Administration of medication infusions	10.06%	21.01%	32.83%	25.16%	9.43%	1.51%	2.03
24. Change of wound dressings	12.83%	38.74%	24.65%	16.60%	4.91%	2.26%	1.61
25. Preparation for test and therapies	14.09%	48.18%	22.77%	9.18%	3.90%	1.89%	1.39
26. Keep patient waiting who rung	18.62%	39.50%	18.36%	14.21%	7.17%	2.14%	1.51
27. Adequate hand hygiene	20.38%	46.67%	18.36%	8.93%	3.65%	2.01%	1.27
28. Necessary disinfection measures	8.43%	59.75%	18.99%	8.68%	2.89%	1.26%	1.37
29. Studying care plans	7.80%	26.67%	30.19%	24.40%	9.31%	1.64%	2.01
30. Assessment of newly admitted patient	9.18%	22.89%	36.73%	21.64%	7.30%	2.26%	1.95
31. Set up care plans	10.19%	27.55%	30.31%	22.77%	7.55%	1.64%	1.9
32. Documentation and evaluation of the care	7.17%	29.81%	32.45%	20.88%	8.30%	1.38%	1.93

regression parameter is 0.004, showing that every "point of depersonalization" raises the BERNCA-R score by an average of 0.004 points. The R^2 coefficient for this model was 24.94%, indicating that 24.94% of the variance in the BERNCA-R outcome was explained by the variables included in the model. The remaining 75.06% depends on the variables not accounted for in the model and random factors. These specific data are presented in Table 6.

Discussion

Level and types of nursing care rationing

The main topic of the work concerned determining the determinants of job satisfaction affecting the rationing of nursing care. In our study, using the BERNCA-R questionnaire, it was established that the phenomenon of rationing of nursing care existed in the surveyed group of nurses, with an average total BERNCA-R score of 1.65 out of 4 possible (SD = 0.81). This score suggests that the frequency of rationing care fell between "never" and "rarely." Similar results in terms of the average BERNCA-R score around 1.65 points (SD = 0.81) on a 0–4 scale were obtained by many other researchers [13, 23, 24]. The study by Schubert et al. [25] closely matched our findings, with a score of 1.69 (SD = 0.57). In Saqer et al.'s study [26], the average level of rationing of care was 2.78 points (on a 1–5 scale).

Schubert et al. [27] in a study on the rationing of nursing care and its correlation with patient outcomes showed that nurses rarely omitted any of the 20 tasks listed in the BERNCA-R questionnaire within 7 days. The average level of implicit rationing of nursing care was 0.82 (SD = 0.26). Papastavrou et al. [28] demonstrated nursing care rationing at a level of 0.83 points (SD = 0.52) using the BERNCA-R questionnaire.

Table 4 Impact of sociodemographic and work-related variables on care rationing

Variables		BERNCA-R [score]			
		M ± SD	Median	Q1-Q3	p-Value
Age	23–30 years (*N* = 134)	1.64 ± 0.77	1.53	1.12–2.09	0.3*
	31–40 years (*N* = 86)	1.65 ± 0.62	1.59	1.17–2.02	
	41–50 years (*N* = 331)	1.7 ± 0.83	1.61	1.15–2.19	
	> 50 years (*N* = 242)	1.59 ± 0.87	1.44	1.01–2.08	
Sex	Female (*N* = 763)	1.64 ± 0.81	1.56	1.09–2.09	0.443**
	Male (*N* = 16)	1.79 ± 0.8	1.75	1.11–2.31	
Education	Medical secondary school (*N* = 328)	1.71 ± 0.83	1.56	1.2–2.23	0.05*
	Medical study (*N* = 89)	1.72 ± 0.84	1.66	1.16–2.08	
	Bachelor of Science in Nursing (*N* = 226)	1.64 ± 0.74	1.56	1.09–2.06	
	Master of Science in Nursing (*N* = 142)	1.5 ± 0.83	1.41	0.96–2.07	
Professional seniority	< 5 years (*N* = 143)	1.64 ± 0.75	1.56	1.14–2.08	0.361*
	6–10 years (*N* = 61)	1.79 ± 0.77	1.75	1.16–2.2	
	11–15 years (*N* = 40)	1.75 ± 0.79	1.69	1.26–2.14	
	16–20 years (*N* = 82)	1.71 ± 0.87	1.66	1.16–2.06	
	> 20 years (*N* = 457)	1.61 ± 0.83	1.5	1.03–2.12	
Specialization	No (*N* = 567)	1.65 ± 0.8	1.56	1.12–2.06	0.879**
	Yes (*N* = 228)	1.65 ± 0.85	1.59	1.03–2.25	
Qualification courses	No (*N* = 456)	1.71 ± 0.79	1.6	1.22–2.16	0.005**
	Yes (*N* = 339)	1.56 ± 0.83	1.47	1–2.06	
Specialized courses	No (*N* = 366)	1.72 ± 0.79	1.59	1.22–2.19	0.013**
	Yes (*N* = 429)	1.59 ± 0.82	1.5	1–2.06	
Other forms of postgraduate education	No (*N* = 711)	1.66 ± 0.81	1.56	1.12–2.12	0.25**
	Yes (*N* = 84)	1.56 ± 0.82	1.44	1.02–2.1	
Number of jobs	1 (*N* = 453)	1.68 ± 0.8	1.59	1.16–2.09	0.154*
	2 (*N* = 234)	1.61 ± 0.79	1.56	1.06–2.19	
	> 3 (*N* = 79)	1.52 ± 0.95	1.42	0.78–2	
Main place of work	Inpatient treatment (*N* = 740) - A	1.69 ± 0.79	1.56	1.13–2.19	< 0.001***
	Outpatient treatment (*N* = 38) - B	1.15 ± 0.76	1.29	0.7–1.78	A > B > C
	Other(*N* = 15) - C	0.57 ± 0.68	0.44	0–0.89	
System of work	Single shift (*N* = 126) - A	1.47 ± 0.91	1.41	0.84–1.87	< 0.001***
	Double shift (*N* = 650) - B	1.7 ± 0.78	1.6	1.16–2.15	B > A,C
	Other (*N* = 14) - C	1.11 ± 0.9	1.17	0.17–1.73	
Average number of patients under care	1–5 patients (*N* = 111) - A	1.35 ± 0.65	1.29	0.96–1.75	< 0.001***
	6–15 patients (*N* = 239) - B	1.56 ± 0.74	1.56	1.04–1.96	D > C,B, A E > B,A C, B > A
	16–25 patients (*N* = 210) - C	1.67 ± 0.78	1.59	1.25–2.03	
	26–35 patients (*N* = 74) - D	2.02 ± 1.01	1.97	1.29–2.88	
	> 35 patients (*N* = 159) - E	1.77 ± 0.85	1.62	1.17–2.34	

Notes: * Kruskal-Wallis test; ** Manna-Whitney test; *** Kruskal-Wallis test + post-hoc analysis (Dunn test)

Abbreviations: BERNCA-R – Basel Extent of Rationing of Nursing Care revised

Influence of sociodemographic and work-related factors on nursing care rationing

In our study, one of the most frequently rationed activities was providing emotional or psychosocial support to patients. Patient satisfaction is significantly influenced by the nurse-patient relationship, respecting their dignity, and assistance in daily activities [29]. Al-Kandari et al. [30] listed activities most commonly incomplete on all wards, such as patient and family communication at 26.8%, nursing care documentation at 23.3%, and oral hygiene at 17.7%. Interpersonal relationships between patient and nurse have been the subject of several studies emphasizing that assessing satisfaction is an indicator of healthcare quality from which service levels can be determined [31].

In our study, the least rationed activities were related to the administration of prescribed medication or infusion at the right time, differing from the Schubert study [25]. This, among other things, illustrates the differences between research findings. In fact, the authors of the study share the observation that nurses have developed an informal hierarchical system to prioritize necessary nursing care when resources are too limited, where monitoring and administering drugs have higher priority and

Table 5 Correlation between SWLS, JSS, LOT-R, and MBI and care rationing

Variables	Spearman's correlation coefficient	*p*-Value
SWLS and BERNCA-R	-0.126	$p < 0.001$
JSS and BERNCA-R	-0.261	$p < 0.001$
LOT-R and BERNCA-R	-0.061	$p = 0.092$
MBI (general score) and BERNCA-R	0.339	$p < 0.001$
MBI: EE and BERNCA-R	0.356	$p < 0.001$
MBI: DEP and BERNCA-R	0.305	$p < 0.001$
MBI: PA and BERNCA-R	0.15	$p < 0.001$

Abbreviations: MBI – Maslach Burnout Inventory General Survey; EE – emotional exhaustion; DEP – depersonalization; PA – lack of personal accomplishments; SWLS – Satisfaction with Life Scale; LOT-R – Life Orientation Test-Revised; JSS – Job Satisfaction Scale; BERNCA-R – Basel Extent of Rationing of Nursing Care revised

are less rationed (similar to other tasks related to patient treatment and safety). Kalisch et al. [2] list timeliness in drug administration as the 6th most frequently missed task. Janikova et al.'s study [32], sought uncompleted interventions in basic patient care, specialized nursing care, and common tasks performed by all those providing patient care, and similar uncompleted activities as in our study were found in each group. In Uchmanowicz et al.'s study [33], rationing activities included patient and family communication, patient monitoring, administration of prescribed medication and/or infusions, ensuring the assessment of newly admitted patients' needs, and assistance to patients experiencing unforeseen sudden or acute changes, because the doctors called were significantly delayed. Hence, most of the presented results converge with our study's findings. It should be noted that some studies obtained different results, especially in terms of the average level of nursing care rationing.

In our study, the tasks least rationed were those related to teaching patients to use the toilet or control their urinary function, hygienic hand washing, and giving attention to disoriented patients. Although rationing these tasks could negatively impact patient health, most of them are related to care and monitoring and could be performed by other professionals, which suggests that if nurses didn't perform these tasks or spent less time on them, as these tasks could be performed by other professional groups, they could dedicate more time to the most frequently rationed tasks, which couldn't be delegated to other professional groups.

Considering the consequences of rationing nursing care in terms of costs, Blackman et al. [9] emphasize that the causes influencing nursing resource conditions lead to a culmination of reasons affecting the occurrence of nursing care shortages. Understanding these relationships is crucial for adopting strategies to minimize the rationing of nursing care. Kalisch et al. [2] also cite regulations from the Center for Medicare and Medicaid Services, indicating that in the event of complications such as pressure ulcers or patient falls, serious consequences for the hospital can occur due to difficulties in obtaining payment for medical services provided.

Among the socio-demographic and work-related characteristics, significant independent predictors of a higher BERNCA-R score (more frequent rationing) included specialization, having 16–25, 26–35, and >35 patients under care. Conversely, among significant independent predictors of a lower BERNCA-R score (less frequent rationing), were work experience of over 20 years and work in open healthcare settings.

Similar to our study in terms of work experience and education influencing the rationing level, yet different in terms of the impact of nurses' age on the rationing level, Al-Kandari et al. study [30] demonstrated varying findings. Nurses more often stated they couldn't document nursing care due to increased workload. Comparing nursing care delivery with education, it was found that nurses with an NC education level more frequently reported not being able to perform tasks during shifts than ADN and BSN nurses. NC-educated nurses more often left incomplete tasks such as urinary catheter care, discharge preparation, but the overall completion of nursing tasks by NC, ADN, and BSN nurses was not statistically significant. Statistically insignificant differences were found in completed tasks compared to nurses' experience, but more incomplete tasks were found among nurses with less professional experience, as commented on by Al-Kandari et al. [30] that time management and work organization are skills that develop with professional experience, hence younger nurses may simply lack these skills, suggesting the need for training related to developing such skills, and it is also important to have nursing teams with more professional experience on duty.

Ausserhofer et al. [34] in a study under the RN4CAST project showed the connection between the work environment and care left undone. Hospitals with a more supportive work environment reported fewer instances of care left undone by nurses. These were hospitals with a lower patient-nurse ratio and a lower percentage of nurses performing tasks not related to nursing care. Blackman et al. [9] in search of factors causing nursing care rationing pointed out that inadequate nursing resources have the greatest impact on why nursing care is reported to be omitted (coefficient 0.61), followed by workload intensity, particularly factors related to an imbalance in patient allocation and inadequate support staff (coefficient 0.43), as well as tensions related to poor team communication. Winsett [35] states that the main cause of insufficient care was the lack of staff, inadequate to the patients' needs, a shortage of auxiliary staff, and sudden events such as patients' deteriorating conditions or admitting severely ill patients, as well as unavailable medications.

Table 6 Results of multivariate regression analysis for the BERNCA-R questionnaire

Item		Parameter	95%CI		*p*-Value
Age	23–30 years	ref.			
	31–40 years	0.145	-0.15	0.44	0.335
	41–50 years	0.301	-0.031	0.633	0.076
	> 50 years	0.321	-0.042	0.684	0.084
Sex	Female	ref.			
	Male	-0.037	-0.427	0.354	0.854
Education	Medical secondary school	ref.			
	Medical study	-0.046	-0.224	0.133	0.615
	Bachelor of Science in Nursing	-0.076	-0.235	0.084	0.353
	Master of Science in Nursing	-0.122	-0.301	0.056	0.18
Professional seniority	< 5 years	ref.			
	6–10 years	-0.121	-0.427	0.185	0.438
	11–15 years	-0.195	-0.563	0.173	0.299
	16–20 years	-0.333	-0.689	0.022	0.067
	> 20 years	-0.485	-0.828	-0.142	0.006 *
Specialization	No	ref.			
	Yes	0.144	0.02	0.268	0.023 *
Qualification courses	No	ref.			
	Yes	-0.033	-0.148	0.082	0.573
Specialized courses	No	ref.			
	Yes	-0.009	-0.123	0.104	0.871
Other forms	No	ref.			
	Yes	-0.101	-0.282	0.081	0.277
Number of jobs	1	ref.			
	2	-0.035	-0.158	0.087	0.573
	> 3	-0.057	-0.248	0.135	0.564
Main place of work	Inpatient treatment	ref.			
	Outpatient treatment	-0.441	-0.689	-0.193	0.001 *
	Other	-0.763	-1.174	-0.352	< 0.001 *
System of work	Single shift	ref.			
	Double shift	0.14	-0.021	0.301	0.089
	Other	-0.102	-0.519	0.315	0.631
Average number of patients under care	1–5 patients	ref.			
	6–15 patients	0.15	-0.027	0.328	0.098
	16–25 patients	0.191	0.005	0.377	0.044 *
	26–35 patients	0.528	0.3	0.756	< 0.001 *
	> 35 patients	0.354	0.155	0.553	0.001 *
SWLS [score]		-0.008	-0.019	0.002	0.119
JSS [score]		-0.007	-0.018	0.005	0.258
LOT-R [score]		0.021	0.006	0.036	0.007 *
MBI: EE [score]		0.006	0.004	0.009	< 0.001 *
MBI: DEP [score]		0.004	0.002	0.006	0.001 *
MBI: PA [score]		0	-0.002	0.002	0.803

Notes: p – multivariate linear regression; * statistically significant relationship ($p < 0.05$)

Abbreviations: MBI – Maslach Burnout Inventory General Survey; EE – emotional exhaustion; DEP – depersonalization; PA – lack of personal accomplishments; SWLS – Satisfaction with Life Scale; LOT-R – Life Orientation Test-Revised; JSS – Job Satisfaction Scale; BERNCA-R – Basel Extent of Rationing of Nursing Care revised

Burnout dimensions and their association with nursing care rationing

In our study, it was shown that a less pessimistic outlook on life, greater emotional exhaustion, and higher depersonalization were significant predictors of nursing care rationing. Similarly, in the study by White et al. [36], as many as 72% of nurses reported one or more unfinished tasks in their last shift due to lack of time or resources, and nurses with burnout were five times more likely to leave care incomplete than nurses without burnout. Uchmanowicz et al. [24] obtained similar results to our own research, demonstrating a statistically significant positive correlation between BERNCA-R and MBI and that one of

the independent predictors of BERNCA-R was the score of emotional exhaustion in MBI.

Perceiving the adequacy of staffing for nursing care tasks was also a significant predictor of satisfaction with current work and profession. This indicates that to increase job satisfaction and patient care, actions related to reducing missing care and ensuring adequate nursing staffing are necessary. Blackman et al. [9] demonstrated that the level of satisfaction of surveyed nurses with their work had an impact on the reasons for lack of care. Bekker et al. [37] proved that job satisfaction was most strongly correlated with uncompleted nursing tasks. Kalisch et al. [38] demonstrated that nursing staff who perceived fewer deficiencies in nursing care and unfinished care in their ward were more satisfied with their current position and profession. Similar results in terms of moderate job satisfaction for nurses were shown by Lorber [39], and Uchmanowicz et al. [24], with an average score of 19.74 ± 5.57 for JSS and a statistically significant negative correlation between BERNCA-R results and JSS scores. Therefore, the demonstrated influence of job satisfaction on the frequency of nursing care rationing implies that, following Sansoni et al. [40], greater attention should be paid to tasks, organizational policy, and professional advancement, as the study showed nurses' dissatisfaction with these aspects of work.

Partially similar results to our own research (mainly concerning nurses dissatisfied with life and work) were obtained by Jaworski et al. [23]. Nurses in the "pessimistic and dissatisfied with life and work" group achieved significantly higher scores on the BERNCA-R scale than those in the "optimistic and satisfied with life and work" group. In the study by Uchmanowicz et al. [41], using BERNCA-R, SWLS, and LOT-R, for SWLS: low life satisfaction – 21.82%; moderate life satisfaction – 37.86%; high life satisfaction – 40.25%. Regarding LOT-R results: high pessimism – 24.36%; average optimism – 41.52%; high optimism – 34.11%. More pessimistic nurses, with low and moderate levels of life satisfaction and neutral life orientation, presented significantly higher scores on the BERNCA-R scale than more optimistic nurses with high life satisfaction. The results presented regarding the influence of life satisfaction on the BERNCA-R score partly correspond to the findings in our own study, indicating that higher life satisfaction leads to less frequent nursing care rationing. However, in terms of the influence of life orientation on the BERNCA-R score, the results differed from those obtained by Uchmanowicz et al. [41]. Firstly, in our study, there were more individuals with a tendency toward pessimism. Secondly, the LOT-R and BERNCA-R (in our study) showed a statistically insignificant relationship. As Uchmanowicz et al. [41] suggest, the question needs to be asked: "Is dispositional optimism more associated with someone's effort, or perhaps with their own interpretation of unfinished nursing care, in which inadequate care is more positively evaluated by the optimist and therefore ignored as an obvious shortcoming?" Therefore, it is necessary to weaken factors and independent predictors that increase the BERNCA-R score, for example, by taking action to reduce emotional exhaustion and stress by promoting assertiveness, relaxation techniques, interpersonal communication [42, 43], supervision, mindfulness-based stress reduction training [44], and conversely, strengthen factors and independent predictors that decrease the BERNCA-R score.

It is important to highlight that the multivariate linear regression model explained approximately 24.94% of the variance in the BERNCA-R score. While this represents a substantial portion of explained variance, it also indicates that about 75% of the variation is accounted for by factors not included in our analysis. These might include variables such as specific organizational policies, individual coping mechanisms, leadership styles, interpersonal team dynamics, or patient characteristics. Future research should therefore explore these additional potential determinants to achieve a more comprehensive understanding of the factors influencing nursing care rationing.

The demonstrated impact of life satisfaction, job satisfaction, and burnout on the level of nursing care rationing indicates the need to promote attitudes of kindness in both patient and colleague relationships. In the study by Papastravrou et al. [45], the analysis showed that teamwork, leadership and autonomy, as well as patient-related communication, collectively accounted for 18.4% of the variance in nursing care rationing. An additional source of information regarding the phenomenon of nursing care rationing, the most commonly rationed activities, and the search for variables explaining the causes of nursing care rationing may be patient satisfaction surveys (their expectations and assessment of the quality of patient care), as conducted by Kalisch et al. [46].

Unexpected findings regarding life orientation

An unexpected finding of our study was the positive association between a less pessimistic (more optimistic) life orientation and higher nursing care rationing scores. This result appears counterintuitive, as optimism is typically considered a protective factor against occupational stress. One possible interpretation is that nurses with a more optimistic outlook might perceive rationed tasks as less critical, underestimate their importance, or experience less urgency in addressing omissions. Additionally, optimists may be more likely to prioritize tasks differently or to assume that omitted care will be subsequently compensated for by others or in future shifts.

Study limitations

This study has several limitations that should be acknowledged. First, it was conducted in a single clinical center, limiting generalizability to other healthcare institutions or regions. Future studies should incorporate multicenter designs and diverse clinical settings to improve external validity and generalizability. Second, self-administered questionnaires may introduce response biases such as social desirability or recall bias. This could be mitigated in future studies by supplementing self-reported measures with observational data, structured clinical assessments, or third-party evaluations. Additionally, due to the cross-sectional study design, causal inferences cannot be established. Prospective, longitudinal research designs could help clarify causal relationships among variables. Lastly, certain factors, including patient characteristics and specific organizational supports, were not fully controlled or measured, potentially influencing results. Further research should aim to comprehensively account for these additional variables.

Implications for nursing practice

The value of the conducted study lies in the fact that studying nursing care rationing in the Polish context is a new direction in research. Therefore, engaging in this issue broadens knowledge about the course of this phenomenon in Poland. The results of this study may encourage further research, exploring this phenomenon in other countries and utilizing experiences in creating work conditions like Magnet Hospitals. The conclusions from this study can be utilized in creating a nurse-friendly work environment, as all necessary tools to limit the causes of nursing care rationing are available to every employer. The nursing profession is independent, self-regulating, and holds public trust. Independence means responsibility, so the nursing environment should strive to minimize the phenomenon of nursing care rationing at all possible levels: scientific, employee-related, and professional self-government.

Policy and practical implications

The results of this study provide critical insights for developing and implementing evidence-based workplace policies and targeted occupational health interventions to reduce burnout and nursing care rationing. Healthcare organizations should focus on optimizing workloads, ensuring adequate staffing ratios, and promoting supportive working environments, particularly within closed healthcare settings and shift-based systems. It is recommended to routinely assess burnout levels among nursing staff, specifically monitoring emotional exhaustion and depersonalization, and to establish proactive measures including stress management programs, mindfulness-based stress reduction training, resilience building workshops, and psychological support systems. Additionally, fostering a workplace culture that promotes job and life satisfaction through recognition, career development opportunities, and flexible scheduling could significantly mitigate care rationing.

Conclusions

A high level of life satisfaction and job satisfaction contributes to a reduction in nursing care rationing. Specifically, higher levels of burnout—particularly in the dimensions of emotional exhaustion and depersonalization—are significantly associated with more frequent nursing care rationing. Additionally, nursing care rationing is more prevalent among staff working in closed healthcare settings and those working in shift-based systems. A higher number of patients under care during a shift is associated with higher levels of burnout and more frequent nursing care rationing; however, given the cross-sectional nature of this study, these findings should be interpreted cautiously and confirmed in prospective longitudinal research.

Abbreviations

BERNCA-R	The Basel Extent of Rationing of Nursing Care Revised
JSS	Job Satisfaction Scale
LOT-R	Life Orientation Test
MBI	Maslach Burnout Inventory
SWLS	Satisfaction with Life Scale

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Author contributions

Dorota Milecka: Conceptualization, Methodology, Data curation, Investigation, Formal analysis, Writing – review & editing, Writing – original draft, Supervision, Project administration, Final approval. Henryka Homętowska: Conceptualization, Methodology, Writing – review & editing, Final approval. Stanisław Manulik: Conceptualization, Data curation, Investigation, Writing – original draft, Final approval. Adrianna Frydrysiak-Brzozowska: Conceptualization, Methodology, Writing – review & editing, Final approval. Izabella Uchmanowicz: Conceptualization, Methodology, Writing – review & editing, Writing – original draft, Funding acquisition, Project administration, Supervision, Final approval.

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Data availability

The datasets generated and/or analyzed during the present study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Bioethics Committee of Wroclaw Medical University (approval no. KB-691/2017). The study adhered to the principles of the Helsinki Declaration and Good Clinical Practice. All participants provided written informed consent to participate in the study.

Consent for publication

Not applicable.

Competing interests
The authors declare no competing interests.

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